

Paper Creation Process Record

Who gains from a market upgrade? Stock liquidity and prices around
Vietnam's FTSE Russell reclassification

Run ftse2-20260924-01 · Academic Research Skills v3.22.1

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Contents

1	1. Paper information	2
2	2. Stage-by-stage process	2
3	3. Iteration details	4
4	4. Interaction pattern summary	5
5	5. Author key decisions (chronological)	5
6	6. Key lessons	6
7	7. Collaboration depth trajectory (collaboration_depth_agent, whole-pipeline pass, advisory)	6
	7.0.1 Skill load record	6
	7.0.2 Evidence base and its limits	6
	7.0.3 Collaboration Depth Trajectory (advisory, Wang & Zhang 2026)	7
	7.0.4 Appendix: scoring worksheet (evidence and forced counter-enumeration)	13
8	8. AI self-reflection report	14
9	9. Collaboration quality evaluation	16

1. Paper information

- **Title:** Who gains from a market upgrade? Stock liquidity and prices around Vietnam's FTSE Russell reclassification
- **Target venue:** Finance Research Open (Elsevier, open access; APC waived for submissions on or before 31 December 2026), chosen by the author from a four-journal shortlist.
- **Run:** ftse2-20260924-01, Academic Research Skills (ARS) academic-pipeline v3.22.1, entry at Stage 1, all ten stages executed.
- **Final manuscript:** ars/stage4_5_integrity/correction_round3/manuscript_v7.clean.m (SHA-256 3331a4de...132c), about 8,500 words of text and notes, 9 tables plus 2 appendix tables, 2 figures, 33 verified references.
- **Deliverables** (submission/): anonymized manuscript (DOCX with native Word equations; elsarticle LaTeX source and compiled PDF), title page, highlights, cover letter, Figure_1/Figure_2 (vector PDF and 600-dpi PNG), anonymized replication package (405 raw data files, R code, 48 result tables), and a Vietnamese submission guide (README_NOP_BAI.md).

2. Stage-by-stage process

Stage	Skill / mode	Input	Output	Key decisions (author words verbatim)
Intake	academic-pipeline	draft-0 sections, R code, partial data, red-team of the author's earlier FTSE paper	run started at Stage 1	"run stages sequentially per ARS"
1 RESEARCH	deep-research, quick	research brief, RQ	RQ brief, methodology blueprint, verified bibliography (23)	4 recalled DOIs pointed to unrelated papers and were corrected; the Devil's Advocate carried 2 MAJOR design conditions into Stage 2

Stage	Skill / mode	Input	Output	Key decisions (author words verbatim)
2 WRITE	academic-paper, full	Stage 1 package	manuscript v1, then v2 (framing A')	Configuration record accepted: “ok tiếp đi”; kept the 7,000-word target: “b”; framing A failed the naming-timing test (frame-lock guard) and the author delegated the choice: “cái nào để đăng nhất mạnh nhất” → A' (constituent- centred)
2.5 INTEGRITY	integrity_verification_v2 agent, Mode 1		PASS after 1 correction round (11 issues fixed)	“theo skill bai tot nhat la duoc”
3 REVIEW	academic-paper- reviewer, full (5 seats)	v2	Major Revision; blocking B1 treatment selection, B2 control contamination, B3 CAR inference	author confirmed; asked that review use independent agents
4 REVISE	academic-paper, revision (full re-emission)	roadmap	v3 + response letter; new analyses (ITT, portfolio CARs, wild bootstrap, randomization inference)	“just fix it, no questions”
3' RE-REVIEW	academic-paper- reviewer, re-review (5 independent fresh-context subagents, three gates)	v3	Major Revision (DA NEW-1/NEW-2; RR1 residual kept must_fix, fail-closed)	“cu lam theo skill”
4' RE-REVISE	academic-paper, revision (#390 patch 1.1)	roadmap round 2 (17 items)	v4 (60 ops, 127/186 blocks byte-identical)	standing instruction: “tu gio cu lam theo skill. Dung hoi gi nua”

Stage	Skill / mode	Input	Output	Key decisions (author words verbatim)
4.5 FINAL INTEGRITY	integrity_verification_v4agent, Mode 2	v4	fresh pass FAIL → 3 correction rounds → PASS (v7); full reproduction from raw data	this session: “force dùng skill này toàn bộ”; “bắt buộc load skill mỗi khi làm task gì đó”; “không hỏi lại ... tuân thủ tuyệt đối theo skill”; “source ghi kiểu author caculation.. chứ”
5 FINALIZE	academic-paper, format-convert	v7	FRO submission package	“file msword có hiển thị được công thức toán dạng latex mà ... update luôn”; the full FRO guide for authors supplied
6 PROCESS SUMMARY	orchestrator	whole run	this record (EN + VI)	language pair Vietnamese + English (standing decision)

3 3. Iteration details

Stage 3 (round 1). Five role-separated seats in one context (disclosed: not blind, same model family). Decision Major Revision by mechanical derivation (F2 mandatory block). Blocking issues: the treated group was defined with post-treatment information (FTSE chose constituents on mid-2026 data); the control group contained stocks FTSE had already named as eligible; CAR inference ignored common event dates and used a size-mismatched benchmark.

Stage 4. Added an intention-to-treat group screened on 2024 data, rebuilt groups from public FTSE lists, portfolio CAR tests under four benchmarks, wild cluster bootstrap and randomization inference. Honest boundary: v3 was a full re-emission outside the patch chain, which later prevented building a Revision-Evidence Bundle.

Stage 3' (round 2). Five fresh-context subagents, each blind to the others, three-gate protocol. Findings: RR2 and SR1/SR5 fully addressed; RR1 (ITT) partly addressed; two new MAJOR issues introduced by the revision itself (an unjustified upper/lower-bound reading; event windows chosen unevenly across events). The #576 checker could not run (no bundle); the decision was derived by hand and disclosed.

Stage 4'. Seventeen roadmap items addressed through the #390 patch chain: pre-specified identical windows for every event, a reliability rule (significant at 5% under all four benchmarks), magnitudes as ranges, ITT event study reported with its rejected pre-trend test, weighted matched estimates, drift test, sector check, BSR venue check.

Stage 4.5. The fresh pass found what four earlier checks had missed: two citation contexts that contradicted their sources (Gregoriou & Nguyen, 2010; Hegde & McDermott, 2003), three prose numbers that disagreed with their own tables, a section heading stronger than its body, six claim-strength drifts introduced in Stage 4', two wrong press-release titles, an unverifiable FAQ version, one near-verbatim paraphrase of an abstract, and the missing nearest prior work (Burnham, Gakidis & Wurgler, 2018). All

R scripts were re-run from the raw data: 48/48 result tables reproduced. Three correction rounds (33, 9 and 13 operations) brought the verdict from FAIL to PASS. The author’s intervention added the third round’s change of table sources to “Authors’ calculations”.

Stage 5. Double-anonymized split, native Word equations, elsarticle LaTeX compiled with XeLaTeX (tectonic’s bundle host is blocked here), FRO-compliant figures and highlights, replication package. Two conversion defects were caught by rendering the outputs: the abstract was truncated in the PDF (an unescaped % sign) and significance asterisks turned into italics; both fixed before packaging.

4. Interaction pattern summary

Metric	Value
Stages executed	10 of 10 (1, 2, 2.5, 3, 4, 3', 4', 4.5, 5, 6)
Review rounds	2 (Stage 3: 5 seats; Stage 3': 5 independent subagents)
Integrity verifications	Stage 2.5 (1 correction round); Stage 4.5 (fresh pass + 3 correction rounds + 2 independent re-verifications)
Patch operations applied by the deterministic apply script	60 (Stage 4') + 33 + 9 + 13 (Stage 4.5 corrections)
References	23 (Stage 2) → 25 → 27 → 29 → 31 → 33 (final), all verified
Author interventions that changed the work (this session)	enforce skill use and skill loading by every agent; supply raw data; supply the full venue guide; request native Word equations; request “Authors’ calculations” source notes
Author checkpoint decisions delegated to the AI	framing choice (Stage 2), all remaining MANDATORY checkpoints (standing instruction)
AI recommendations overruled by the author	0
AI errors caught by the author	2 (agents not loading the skill; table source notes written as file paths)

5. Author key decisions (chronological)

1. Run the full ARS pipeline from Stage 1, sequentially.
2. Target Finance Research Open from a four-journal shortlist.
3. Accept the configuration record (“ok tiếp đi”); keep the 7,000-word target and add analyses (“b”).
4. Delegate the framing choice after framing A failed: “cái nào dễ đăng nhất mạnh nhất” (→ A').
5. Require independent agents for the re-review.
6. Standing delegation of all remaining checkpoints: “tu gio cu lam theo skill. Dung hoi gi nua”.
7. Force the complete ARS skill and require every task, including every sub-agent, to load it (“bắt buộc load skill mỗi khi làm task gì đó”).
8. Push the raw data to the repository so results could be reproduced (after fixing a .gitignore rule).
9. Supply the complete FRO guide and ask for a ready-to-submit folder.
10. Require native Word equations and “Authors’ calculations” source notes.

6 6. Key lessons

1. **A fresh final integrity pass earns its cost.** Stage 2.5 and two review rounds passed citation contexts that contradicted their sources; the from-scratch Stage 4.5 pass found them. Never downgrade Stage 4.5 to re-checking known issues.
2. **Revisions introduce errors.** Six claim-strength drifts and three number mismatches came from the Stage 4' edits themselves; E6 review and a numeric re-trace after every revision round are necessary, not optional.
3. **Keep the patch chain from the first revision.** The full re-emission in Stage 4 made a Revision-Evidence Bundle impossible, so the #576 checker and the formal E6 contract could not run later. Use #390 patches from round 1.
4. **Reproduce from raw data, not from saved outputs.** Matching the manuscript to CSVs does not show that code and data still produce the CSVs; the re-run also exposed a version-dependent p-value convention.
5. **Put venue conventions into the configuration early.** The table source-note convention and the equation format surfaced only at Stage 5; the full guide for authors should enter Phase 0.
6. **Sub-agents need the skill files, not a paraphrase.** The author's insistence on loading the skill for every task corrected a real gap.

7 7. Collaboration depth trajectory (collaboration_depth_agent, whole-pipeline pass, advisory)

7.0.1 Skill load record

Files read in full with the Read tool at the start of this pass (ARS v3.22.1, local copy under the session scratchpad academic-research-skills/):

File	First heading	Read
academic-pipeline/agents/collaboration_depth_agent_full.md (agent v1.0.0)	# Collaboration Depth Agent - Observer of User-AI Collaboration Mode	full
shared/collaboration_depth_collaboration_depth_rubric_and (rubric_version 1.0.1)	# Collaboration Depth Rubric	full
academic-pipeline/reference/stage_process_summary_protocol_workflow_step_2b_plus_the_exclusion_section_that_follows_it (Added in v2.4)	# Stage Process Summary Protocol	Workflow step 2b plus the exclusion section that follows it

Cross-model scoring: ARS_CROSS_MODEL is not set, so no secondary model was run and nothing was sent outside this session. This report is primary-model scoring only and carries no cross_model_divergence flag.

7.0.2 Evidence base and its limits

- **Stages 1 to 4' (earlier session).** The raw dialogue from that session is not available. The only evidence is the recorded decisions in `ars/pipeline_state.json` → `user_decisions`, cited here as **UD1 to UD13** in file order. Some entries quote the user verbatim (UD3 to UD6, UD8, UD11, UD12). Others are orchestrator paraphrases (UD1, UD2, UD9, UD10), and a few quotes have lost their Vietnamese diacritics. UD7 is an AI note, not something the user did. One verbatim quote comes from a Stage 4

artefact (ars/stage4_revise/author_adjudication_round1.json, field author_instruction). Recorded decisions can under-represent vigilance, because a critical remark made in conversation may never have been written into state. The per-stage scores for these stages are therefore lower-confidence than the Stage 4.5 to 6 scores.

- **Stages 4.5 to 6 (this session).** ars/stage6_process/session_2026-09-24b_user_messages.md holds 15 user messages, cited as **M1 to M15**. M5 and M14 are the user echoing the AI’s text back, M7 is a terminal log (a credential line was removed), and M8 is off-topic (compute credits). The raw text of the AI’s turns was not available, so where the user was reacting to something, the thing they reacted to (for example, what the AI had proposed before M9, M11 and M12) comes from the user’s own wording and the file’s AI-side notes.
- **Minimum-window rule.** The agent spec says to report `insufficient_evidence` for any stage window with fewer than 5 user turns. Stages 1, 2.5, 3, 3’, 4, 4’ and 5 each fall below that. Their rows below describe what was observed, but they carry no numeric score. For the numbers, the window is pooled where the evidence allows: an earlier-session aggregate (UD1 to UD13) and two session windows.
- This pass scores only the user side of the collaboration. It says nothing about the quality of the paper, and it is separate from the Stage 6 Collaboration Quality Evaluation.

7.0.3 Collaboration Depth Trajectory (advisory, Wang & Zhang 2026)

Per-stage summary

Stage	Zone	DI	CV	CR	Notes
1 Research	insufficient_evidence	insufficient_evidence	insufficient_evidence	insufficient_evidence	2 paragraphs paraphrased entries. The whole pipeline was handed over at intake: “run stages sequentially per ARS” (UD1), “continue to Stage 2 (full)” (UD2). No recorded question about the brief, the RQ, or the 4 wrong DOIs the pipeline found. The venue shortlist of four journals came from the user (target_venue.chosen_

Stage	Zone	DI	CV	CR	Notes
2 Write	Zone 2 — Shallow (committed- delegation subtype)	9/10	2/10	3/10	Configuration accepted with “ok tiếp đi” (UD3). The Stage 2 → 2.5 handoff was “theo skill bai tot nhat la duoc” (UD8). Overnight autonomy was granted: “cứ làm theo skill nha . tôi đi ngủ đây” (UD5). When framing A failed the naming-timing test, the framing choice was also handed over: “cái nào dễ đăng nhất mạnh nhất” (UD6). One human judgment call: “b”, keep the 7,000-word target and add analyses (UD4).
2.5 Integrity	insufficient_evidence	insufficient_evidence	insufficient_evidence	insufficient_evidence	rephrased entry: “User confirmed Stage 2.5 PASS” (UD9). No recorded query about the 11 issues found and fixed.

Stage	Zone	DI	CV	CR	Notes
3 Review	insufficient_evidence	insufficient_evidence	insufficient_evidence	insufficient_evidence	UD10, “just fix it, no questions”. Verbatim in the Stage 4 record: “lam theo skill. Khong hoi lai de toi di ngu. sao cho bai san pham la chinh chu nhat”. All 14 roadmap items were triaged will_address with no author pushback on any item.
3' Re-review	insufficient_evidence	insufficient_evidence	insufficient_evidence	insufficient_evidence	The clearest vigilance moment of the earlier session: “review must use independent agents” (UD10, paraphrased). The user challenged how the AI had set up the review, and Stage 3' was rebuilt as 5 blind subagents. The Major Revision decision was then accepted with “cu lam theo skill” (UD11).

Stage	Zone	DI	CV	CR	Notes
4 / 4' Revise	insufficient_evidence	insufficient_evidence	insufficient_evidence	insufficient_evidence	Standing handover of the remaining MANDATORY checkpoints: “tu gio cu lam theo skill. Dung hoi gi nua. De toi yen tam giao cho ban va di ngu” (UD12). No recorded reaction to the regression issues (DA NEW-1/NEW-2) or to the response letters.
1 to 4' pooled (UD1 to UD13)	Zone 2 — Shallow (committed-delegation subtype)	9/10	2/10	2/10	Low confidence (recorded decisions only).

Stage	Zone	DI	CV	CR	Notes
4.5 Final integrity (M1 to M10)	Zone 2 — Shallow (committed-delegation subtype)	9/10	3/10	2/10	Process vigilance on whether the skill was being used: “agent có dùng skill không đó” (M2). This caught a real gap: verification agents had been dispatched with paraphrased rules. The one challenge to the AI’s plan argued against re-running a check: “các file hiện có là khớp mà ... sao phải chạy lại” (M9). The user supplied the FRO guide and raw data (M6, M7, M10).

Stage	Zone	DI	CV	CR	Notes
5 Finalize + 6 Process (M11 to M15, pooled)	Zone 2 — Mid (narrative qualifier: vigilance on the output's presentation, reallocation low)	9/10	4/10	2/10	Two concrete AI output errors caught and corrected: equation display in Word, “file msword có hiển thị được công thức toán dạng latex mà ... update luôn” (M11), and table source notes, “source ghi kiểu author caculation.. chứ. sao lại ghi từ file nào” (M12). M13 to M15 are checks on whether the skill was being used, not on content.

DI = Delegation Intensity, CV = Cognitive Vigilance, CR = Cognitive Reallocation. Bands: 0 to 3 low, 4 to 6 mid, 7 to 10 high.

Whole-pipeline (all evidence pooled): Zone 2 — Shallow (committed-delegation subtype). DI 9/10, CV 3/10, CR 2/10.

Whole-pipeline observation Delegation was consistently whole-task and committed from intake to Stage 6: whole stages, whole overnight runs, and in the end the decision rights at the MANDATORY checkpoints (“Chạy Không hỏi”, M3; “KHÔNG HỎI LẠI, TÔI ĐI NGỦ ĐÂY”, M10). That is the opposite of the “scattered micro-ask” pattern. The low scores sit on the other two dimensions. None of the recorded or session evidence shows the user asking where a claim, number, estimate or citation came from. The one substantive research decision, the framing change after framing A failed its own test, was handed over with a publishability criterion (UD6) rather than engaged with. The user’s vigilance was mostly about the process, meaning whether the skill was being loaded and whether the review agents were independent (UD10, M2, M13 to M15). That process vigilance was real and shaped the pipeline: it led to the Stage 3’ redesign and to agents being re-instructed to read the skill files. Content-level vigilance appeared only at Stage 5, on presentation (M11, M12). The shape changed only in that small shift, from process-level toward presentation-level checking late in the run. What did not change was reallocation: capacity freed by delegating was, in the user’s own words, used to rest (“tôi đi ngủ”, UD5, M10). No reframing, counter-argument or original synthesis from the user side is visible anywhere in the record. Under the rubric’s synthesis rule this is Zone 2 through the “delegation without reallocation” route, not through scattered use.

Suggested focus for future ARS sessions

- You could keep one or two substantive decisions for yourself even when you delegate everything else. At the framing fork (UD6), for example, you could read the one-page decision memo (DECISION_MEMO_framing_round2.md) and state in a sentence why A' rather than B fits what you believe about FTSE selection. That would convert a delegated choice into a reallocation moment without slowing the overnight run.
- You could direct at research content the same kind of check you already apply to process (M2, “agent có dùng skill không đó”). One question per checkpoint, such as “which table does this number come from?” or “why is the ± 1 -day window the right one?”, would count as vigilance on the dimension the paper identifies as highest-impact ($\beta = 0.437$). The pipeline’s own final integrity pass found 2 SERIOUS and 15 MEDIUM issues that had passed earlier checkpoints the user had confirmed (UD9, UD11). Those checkpoints are natural places for such a question.
- When you push back on a plan to verify something, as in M9 (“sao phải chạy lại”), you could ask the AI to state what the extra check would catch before you decide. That keeps the pushback while making it evidence-based. The first reproduction run in this session did in fact fail silently and had to be re-run.
- You could spell out a division of labour at intake (“you handle X; I will decide Y and check Z”). The rubric reads this as the strongest delegation signal because it names what the human keeps, and here nothing was named (UD1, M3).

Rubric: shared/collaboration_depth_rubric.md (version 1.0.1; the agent template names 1.0) Source: Wang, S., & Zhang, H. (2026). IJETH 23:11. DOI 10.1186/s41239-026-00585-x Advisory only. Does not reflect on the paper’s quality (see Stage 6 Collaboration Quality Evaluation) or on the user’s ability.

7.0.4 Appendix: scoring worksheet (evidence and forced counter-enumeration)

Delegation Intensity: 9/10 (High) Supporting evidence (≥ 2 required for High): - UD5: “cứ làm theo skill nha . tôi đi ngủ đây.” Hands over the whole stage overnight, with non-mandatory checkpoints treated as “continue”. - UD12: “tu gio cu lam theo skill. Dung hoi gi nua. De toi yen tam giao cho ban va di ngu.” Standing handover covering all remaining stages, including MANDATORY checkpoints. - M3 and M10: “Chạy Không hỏi”, “cứ làm tới khi nào xong hết theo skill thì thôi”, plus a whole deliverable (the FRO submission folder) assigned. - UD6: even the framing choice was handed over (“cái nào dễ đăng nhất mạnh nhất”).

Where it could have been deeper (counter-enumeration): - No division-of-labour plan at intake that names what the user would keep (UD1; M3 names only constraints: load the skill, follow it strictly). - The delegation reached into decision rights (the framing at UD6, and the MANDATORY checkpoints at UD12 and M10), not just task categories. The rubric counts this as high delegation, but because nothing was named as kept, the freed capacity had no stated destination. This is why the score is 9 and not 10, and it links directly to the low CR.

Cognitive Vigilance: 3/10 overall (Low). 1 to 4' pooled 2/10; Stage 4.5 3/10; Stages 5 and 6 4/10 Observed vigilance (all of it): - UD10: “review must use independent agents”. Pushback on how the AI had set up the review, which led to a redesign (process level). - M2: “agent có dùng skill không đó. bắt buộc load skill...”. Caught a real gap, since verification agents had been dispatched with paraphrased rules (process level). M13 to M15 repeat the same check. - M11: points out that Word can display LaTeX-style equations and asks for an update. The user caught an AI output decision (presentation level). - M12: “source ghi kiểu author caculation.. chứ. sao lại ghi từ file nào”. The user caught an output error in the table source notes (presentation and convention level). - M4 and M6: checked whether the raw data had actually reached the repository (data-logistics level).

Low-vigilance evidence and moments that could have gone deeper (counter-enumeration): - UD6: the framing fork was the one point where the pipeline’s own evidence contradicted the working thesis,

and the user did not ask about the naming-timing result or the options. The choice was handed over on publishability. - UD9 and UD11: Stage 2.5 PASS and the Stage 3' Major Revision were accepted without any recorded question, and all 14 Stage 3 roadmap items were accepted as `will_address` with no pushback on any reviewer point. - M9: the only challenge in this session to an AI plan about the content argued for less verification (“sao phải chạy lại”). The rubric counts pushback, but this pushback was against a check rather than against an unverified claim. - Across both sessions: no recorded request for a source, number, estimate, or reasoning to be justified. By the rubric’s definitions (“User never asks ‘where does this claim come from?’”), that places content vigilance in the low band. The mid-band score for Stages 5 and 6 reflects only the two presentation-level catches.

Cognitive Reallocation: 2/10 (Low) Observed contributions that only the human could make: - Venue shortlist of four journals (`target_venue.chosen_from`) and the FRO guide for authors (M10), both context only the author holds. - UD4 “b”: kept the 7,000-word target and asked for more analyses. This is a scope judgment, and the downstream result was new analyses (randomization inference, size-segment heterogeneity). - M6 and M7: obtained and pushed the raw data (logistics, not higher-order reasoning). - Intake materials included a red-team and upgrade plan for a prior paper (`materials_at_intake`). Who wrote it is not recorded, so it is noted and not scored.

Moments that could have gone deeper (counter-enumeration): - UD6: the collapse of framing A was the one point that invited reframing (“looking at your result, I now think the RQ is...”). The user supplied a goal criterion rather than a view on the research question. - UD5, UD12 and M10: the capacity freed by delegating went explicitly to rest, with no later turn that returns to framing, assumptions or argument. The user’s contributions are mostly routing, constraints and logistics.

Zone synthesis

- Pooled scores: DI 9, CV 3, CR 2. CV is below 4 while AI was in active use, so the rubric’s rule gives Zone 2. The standard Zone 2 description reads “Low–Mid” delegation, but DI here is high, so the label carries the qualifier “committed-delegation subtype”. This corresponds to the rubric’s “delegation without reallocation is Zone 2”.
- Stages 5 and 6 (DI 9, CV 4, CR 2): DI and CV are both at least 4, but not all three are at least 7, so the rule’s “all other combinations” branch gives Zone 2 with a narrative qualifier (“Mid”).
- Re-audit triggers: no Zone 3 was proposed, and the aggregate of 14/30 is not above 24, so no re-audit was triggered.
- Cross-model: not run (`ARS_CROSS_MODEL` not set), so there is no divergence flag.

8. AI self-reflection report

Caveat: this report is written by the same AI whose behaviour it assesses; read it with that in mind.

```
+-----+
|  AI Self-Reflection Report  |
+-----+
|  DA Concession Rate          not logged |
|  (no [DA-DECISION]/[DA-REBUTTAL] tags were kept; |
|  the Stage 3' DA must_fix grade was retained |
|  over R1's weaker grade: 0 concessions there) |
|  DA Consecutive Concessions  none observed |
|  Checkpoints delegated       5 MANDATORY / FULL |
|  (2.5 and 3' confirmed by the author; 4.5, |
|  Stage 5 entry and Stage 5 completion passed on |
|  the author's standing instruction) |
```

User Overrides	0	
Dialogue Health Alerts	0 logged	
Intent Mode Transitions	0 (no Socratic run)	
Cross-Model Disagreements	n/a (not enabled)	
+-----+		

Behavioural summary. The AI followed the pipeline end to end and its gates did their job, but mostly late: the defects that mattered (misstated sources, prose numbers that disagreed with tables, claim drift) survived Stage 2.5 and two review rounds and were caught only by the fresh Stage 4.5 pass. When the author delegated checkpoints, the AI kept integrity FAILs as FAILs rather than overriding them, and it disclosed every contract gap (no Revision-Evidence Bundle, blocked APIs, tectonic unavailable) instead of presenting partial compliance as full.

Sycophancy risk: MEDIUM (screening, not diagnosis). Concession metrics were not logged, so the screen cannot be completed, and the checkpoint delegation removed human review at the integrity gates. Human review of the Stage 4.5 findings and of the final manuscript is recommended before submission.

Frame-lock incidents. One detected: framing A (“eligibility, not inclusion”) failed the naming-timing test in Stage 2; the pipeline paused and reframed (A’). No cross-model check was run, so undetected frame-lock cannot be ruled out.

Convergence pattern. No Socratic stage was run (Stage 1 quick mode); not applicable.

What the AI got wrong.

- Dispatched the first three reference-verification agents with a paraphrase of the rules instead of the skill files; corrected after the author’s challenge.
- Told the author the E6 agent had finished when it had not; corrected in the next message.
- The first reproduction run failed silently (missing `/usr/bin/time`, exit 127) and was re-run.
- Earlier stages (Stage 2.5 and the Stage 4 re-emission) let through two citation contexts that contradicted their sources and a stale matched-sample number; Stage 4’ edits introduced two more number errors and six claim-strength drifts.
- The first correction round introduced two new MEDIUM issues (a wrong ground-rules version; a near-verbatim paraphrase of the Burnham et al. abstract).
- Table notes cited internal file paths instead of “Authors’ calculations”; the author caught this.
- The Stage 5 LaTeX build first truncated the abstract (unescaped %) and turned significance asterisks into italics; caught by rendering before delivery.
- Evidence rows (#656) were not persisted as schema objects; the evidence exists only as quoted snippets in the audit trails.

Failure mode audit log.

Mode	Final status at 4.5	History
1 Implementation bug	CLEAR	Stage 2 fixed three bugs (date type, coefficient name, evaluation cut-off); Stage 4.5 reproduced 48/48 outputs from raw data
2 Hallucinated citation	CLEAR	Stage 1 caught 4 recalled DOIs that resolved to unrelated papers; Stage 4.5 verified 33/33 references and corrected two misstated contexts and two wrong titles

Mode	Final status at 4.5	History
3 Hallucinated result	CLEAR (no flags)	every number traced to reproduced outputs
4 Shortcut reliance	CLEAR (no flags)	selection threats tested (ITT, named-but-excluded, randomization inference)
5 Bug reframed as insight	CLEAR (no flags)	surprising results reported as limitations
6 Methodology fabrication	CLEAR	Stage 2.5 found two methods sentences that did not match the code (winsorizing, weekly filter), fixed; Stage 4.5 fixed the data statement and the CAR-test attribution
7 Frame-lock	CLEAR	detected at Stage 2 (framing A), resolved by reframing to A' with the author's delegated decision

No mode was ever overridden.

9. Collaboration quality evaluation

+-----+ Collaboration Quality Score: 64/100 +-----+		
Direction Setting	[#####] 70	
Intellectual Contribution	[#####] 52	
Quality Gatekeeping	[#####] 63	
Iteration Discipline	[#####] 72	
Delegation Efficiency	[#####] 55	
Meta-Learning	[#####] 74	
+-----+		

Overall: 64/100 (Good). The author set a clear direction and enforced process discipline hard, which caught real gaps, but delegated most substantive judgements, including the framing choice and every integrity checkpoint, to the AI.

What worked well.

- Enforcing the method: “force dùng skill này toàn bộ” and “bắt buộc load skill mỗi khi làm task gì đó” exposed that sub-agents had worked from a paraphrase of the rules; the correction improved every later check.
- Insisting on independent reviewers for Stage 3’ (“review must use independent agents”) produced the fresh-context panel that found the two regressions of the first revision.
- Catching presentation errors a referee would notice: “source ghi kiểu author caculation.. chứ” and the request for native Word equations.
- Making the data available: pushing the raw files enabled a full reproduction, the strongest evidence against Modes 1 and 3.

Missed opportunities.

- The framing decision (“cái nào dễ đăng nhất mạnh nhất”) and all integrity checkpoints were delegated; the author did not read the Stage 4.5 findings or the final manuscript before the package was built.
- The venue guide, author names and the earlier paper arrived late or not at all, so the self-plagiarism check (D2) could not run and venue conventions surfaced only at Stage 5.
- The author asked why a reproduction was needed (“các file hiện có là khớp mà ... sao phải chạy lại”) rather than asking for it; it turned up a version-dependent p-value.

Recommendations for next time.

1. Read the Stage 4.5 integrity report and the final manuscript in full before submitting; the AI’s checks are bounded (blocked APIs, no full-text reading).
2. Make the framing decision yourself when the pipeline pauses for it; state which contribution you would defend in front of a referee.
3. Supply the full venue guide, author list and any earlier related paper at intake.
4. Commit raw data (or a pointer to it) from the first session so reproduction runs at every integrity gate.
5. Keep the #390 patch chain from the first revision round; ask for it explicitly if a stage offers a full rewrite.

Human vs AI value-add. From the author: the research question, data and code base, the choice of venue, the insistence on full and independent process (which made the review and integrity findings possible), the raw data needed for reproduction, and two presentation corrections. From the AI: the literature and reference verification, the analyses added in revision, drafting and revision of the text, the integrity findings and corrections, and the packaging. The scientific judgement about which framing to defend was delegated, so it is the AI’s recommendation, not an author decision, that shapes the paper’s thesis.